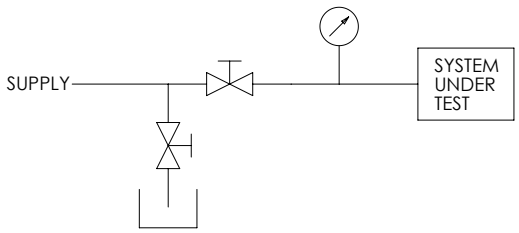
Introduction

BiS Valves offer an extensive range of flow control valve products, in cartridge, manifold mounted and ported installation formats.

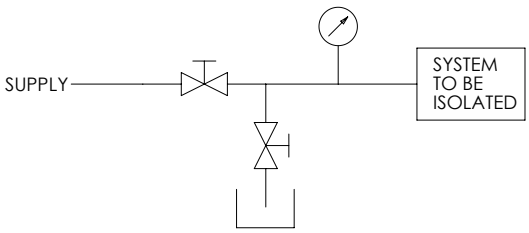
Stop Valves

Stop valves, often known as shut-off or isolation valves, are manually operated valves used to block or open flow lines. Nearly all hydraulic and pneumatic circuits contain stop valves as a critical part of the functionality. Typical applications include isolation of hydraulic circuits or components, providing a method of releasing pressure, and the locking in of pressure during pressure tests.

Typical Uses



When pressure testing it is desirable to minimise the number of leakage points exposed to the system under test. In the configuration shown supply pressure can be removed leaving only trapped pressure in the system being tested.

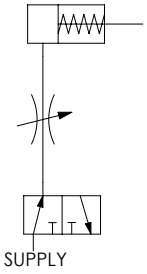


When it is necessary to isolate and remove pressure from part of a system, a block-and-bleed configuration can be used.

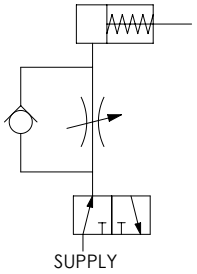
Metering Valves

Metering valves, often known as needle valves, are used to regulate flow and are most often used to control the speed of an actuator. They can be used in either "meter-in" or "meter-out" speed control systems and many are available with integrated check valves such that the flow control is provided in only one direction. Metering valves are not designed to provide a shut-off function and where this is required it is necessary to fit a separate Stop valve.

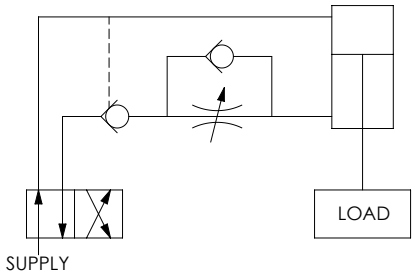
Typical Uses



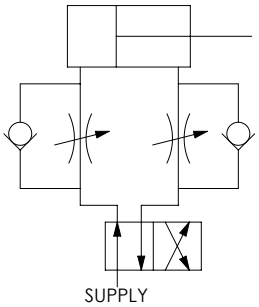
In the configuration shown the metering valve provides bi-directional speed control of the actuator. The speed when extending is likely to differ from the retract speed as it will be dependent on pressure drop across the metering valve.



Using a metering valve with integral check, speed control is provided in the extend direction only. Reversing the connections to the metering valve allows speed control to be in the retract direction only.



When extending the actuator, placing the metering valve on the "meter-out" side of the actuator minimises the risk of a run-away load and associated cavitation. Adding a pilot operated check valve further reduces risk by ensuring that a positive pressure is always present on the full bore side of the actuator.



Utilising two metering valves with integral check valves provides entirely independent speed control of both the extend and retract functions.

Pressure Compensated Flow Regulators

Where either the supply pressure or the pressure required to operate an actuator is variable, the use of standard metering valves can lead to a high variation in actuator speed. In these circumstances, a pressure compensated flow regulator can be used, and this will provide an approximately constant flow rate over a wide range of differential pressures. Pressure compensated valves cannot be used to provide a shut-off function and where this is required it is necessary to fit a separate stop valve.

Selecting the Correct Product

Selection guide for stop valves

	Page	Suitable for liquids	Suitable for gases	1/4" – 3/8" BSP / NPT ports	1/2" – 1" BSP/NPT ports	Medium pressure ports	Cartridge installation	Manifold mount installation	T-bar handle
BNV10FC	5								
BNV25	6 - 7								
BNV25-G	8 - 9								
BNV25-TB	10 - 11								
BNV50	12								
BNV50-G	13								
BNV44AE	14 - 15								
BNV56AE	16 - 17								
BNV81AE	18 - 19								

Important – This is a guide to selecting products that may be suitable. The valve data sheet and application notes must be thoroughly reviewed prior to final selection. Please contact us if the options you require are not shown

Selection guide for metering valves

	Page	Suitable for liquids	Suitable for gases	1/4" – 3/8" BSP / NPT ports	1/2" – 1" BSP/NPT ports	Cartridge installation	Manifold mount installation	Integral check valve	Pressure Compensated
MF25	20 - 21								
FMF25	22								
VFMF25	23								
MF37	24								
MF50	25								
MF75	26								
MF100	27								
CMF25	28								
CMF37	29								
CMF50	30								
CMF75	31								
CMF100	32								
IMC25	33								
IMC37	34								
IMC50	35								
PCFR50	36								

Important – This is a guide to selecting products that may be suitable. The valve data sheet and application notes must be thoroughly reviewed prior to final selection. Please contact us if the options you require are not shown

Application Notes

Intended Use

The products in this catalogue have many potential applications and are used across a wide variety of industries. They are primarily intended for use in industrial and onshore and offshore topside applications. Products should always be used within the parameters specified on the product data sheet, and should be installed, operated and maintained in accordance with the instructions supplied. BiS Valves will offer all possible assistance to ensure correct product selection, but it is the ultimate responsibility of the customer to ensure that the final product selection is suitable for the intended application.

Prohibited Uses

The products in this catalogue are not intended for use in automotive, shipbuilding, aerospace, military or nuclear applications. The products are not designed for contact with foodstuffs. Use with oxygen or with oxygen rich gas mixes is prohibited. Please contact BiS Valves prior to ordering products for use in any of the above applications.

Subsea Applications

Suitability for subsea applications should not be assumed unless it is specifically stated on the product data sheet. In many cases BiS Valves have subsea variants of standard catalogue products available, or can design bespoke subsea variants specifically for your application. Please do not hesitate to contact BiS Valves with your requirement.

Where the product data sheet states suitability for short term subsea use, this should be interpreted as meaning that the product is capable of functioning subsea but that corrosion will occur over time. The acceptable length of deployment will depend on the specific subsea environment and the effect of that environment on the exposed materials of construction, which should be evaluated by the customer.

Where the product data sheet states suitability for subsea use, this should be interpreted as meaning that the product is capable of functioning subsea and that the externally exposed materials of construction will withstand longer term subsea use. The acceptable length of deployment will depend on the specific subsea environment and the effect of that environment on the exposed materials of construction, which should be evaluated by the customer. It is recommended that BiS Valves are consulted where deployment is intended to exceed five years without retrieval and refurbishment and for deployment depths greater than 3,000 metres. For all long term deployments, it is recommended that the system is designed such that failure of a single component does not cause failure of critical system functions.

For subsea applications, take care to choose an appropriate subsea suitable piping system. BiS Valves recommend that particular care is taken if selecting BSP ported products for subsea use.

Cleanliness

Fluid cleanliness is critical to the reliability of all fluid power products. BiS Valves take every precaution to ensure that products are clean at the time of supply. Component parts are cleaned prior to assembly and testing is carried out with test fluids verified as clean to a maximum of AS4059 class 6B-F.

For maximum service life it is recommended that the products are used with fluid maintained at or below AS4059 class 6B-F. Use with higher levels of contaminant should be expected to result in a reduction in service life. Certain product designs are inherently more robust in systems where fluid cleanliness cannot be guaranteed, and it is recommended to contact BiS Valves prior to ordering products for use in such applications.

There is an increased risk of damage to the product at the time of installation, due to contaminant generated during the assembly of the system and associated pipework. Precautions should be taken to ensure that contaminant is not transferred into the product during flushing procedures and initial system operations.

Where fluid contamination presents a risk of failure to critical valve functions, it is recommended to consider fitting a device protection filter from the BiS Valves F or EF ranges. However, this should not be considered as being a substitute for sound hydraulic cleanliness practise.

Product Testing

All products are tested for proof pressure integrity, typically at 1.5 times the working pressure, and are tested to ensure correct functionality as part of our standard manufacturing process. Testing will be carried out using an appropriate test fluid that suits the specific product being tested. The test fluid used will be referenced on the test certificate supplied with the product. In cases where the test fluid used is not compatible with the media in the system into which the product will be installed, the customer may need to flush the product prior to installation. If the product to be supplied must be tested using a specific fluid, please ensure that this is requested prior to quotation and ordering.

Leakage

BiS Valves products are typically designed to provide the very low leakage levels expected by the offshore oil and gas industry, other than certain products such as flow control valves that are not intended to fully close. The actual leakage rate achieved will depend on various factors, such as fluid viscosity and pressure differential, but with liquids is expected to be below a level where it is appropriate to measure by collecting a fluid volume or counting drops, hence a pressure decay and trapped volume method of leakage detection is used. If the product to be supplied must be tested to prove a specific leakage rate, please ensure that this is requested prior to quotation and ordering, such that the suitability of the product and the standard test procedure may be assessed.

Materials
Each product data sheet gives limited guidance relating to the generic type of operating media with which the product is suitable, and the materials of construction. As part of our quotation and order process, BiS Valves will supply an Application Data Request form and from the data supplied will give all possible assistance to ensure correct product selection. However, since it is not possible for BiS Valves to validate products against every conceivable operating parameter, it is essential that the customer satisfies themselves that the product is suitable for their specific application prior to purchase and use.

Alternative construction materials, including exotics, can often be made available on request.

Seals
Each product data sheet contains model code options for various seal materials that are available as preferred options. Many other seal materials and grades are available. Please contact us for a quotation.

Where generic standard materials are specified (eg. Viton®, nitrile, etc), the exact grade of material cannot normally be guaranteed. If you require a specific grade of seal material, please specify when requesting a quotation.

When choosing or specifying a seal material, always ensure that the material is suitable for the application in terms of both temperature and chemical compatibility. Always seek advice from the fluid manufacturer if in doubt with regards to material compatibility. Never specify Ethylene Propylene seals for use with mineral oils or other hydrocarbon based media.

Viton® and Kalrez® are registered trademarks of E.I. du Pont de Nemours and company or its affiliates. BiS Valves use genuine Viton® and Kalrez®.

Pressure Equipment Directive 97/23/EC
The products in this catalogue are classified as Pressure Accessories in terms of the definitions within the Pressure Equipment Directive. Since the products fall below sizes specified within the Directive, the products fall outside categories I to IV, and hence are engineered in accordance with "sound engineering practise" and are not supplied with a CE mark.

ATEX Directive 94/9/EC
The majority of products supplied by BiS Valves can be used in hazardous areas containing potentially explosive atmospheres covered by the ATEX and DSEAR regulations, for zones 1 and 2. As standard, products are not CE marked and should not be used in these environments. However, should use in these areas be required, please ensure that the requirement is specified, along with the zone in which the equipment is to be used, when requesting a quotation or ordering. Following assessment, the products can normally be supplied with a CE mark and declaration of conformity at no extra charge.

NACE MR01-75
Many products supplied by BiS Valves can be configured for use in sour H2S containing environments. This often requires some material changes from the standard product. Where compliance with NACE MR01-75 is required, please specify when requesting a quotation.

Traceability
Unless otherwise requested, only the main pressure containing component in finished products are traceable back to original material. Should either full or a greater level of traceability be required, please ensure that this is specified when requesting a quotation.

Documentation
Standard documentation supplied with BiS Valves products includes a pressure test / conformity certificate, parts list (including available spare parts), assembly drawing, installation, operation and maintenance instructions, and a material certificate for the main pressure containing component, along with any EC conformity declarations appropriate to the product. If further documentation is required, please specify when requesting a quotation. Documenation is delivered as a hard copy with the goods. If electronic documentation is required, please specify when ordering.

Maintenance
Due to the wide variety of applications in which BiS Valves products are utilised, it is impossible to set generic service intervals. It is recommended that the end user should assess their application, including factors such as duty cycle, hostility of the environment both inside and outside the product, and the criticality and consequence of failure, and set in place a preventative maintenance schedule accordingly. It is further recommended that performance is monitored and reviewed to refine the balance between reliability and cost.

It is recommended that products are returned for repair and refurbishment by BiS Valves. However, it is acknowledged that this may not always be possible, and Seal Kits and Repair Kits are available for most BiS Valves products. Product maintenance should only be carried out following safe isolation from the system, in clean conditions, by competent personnel, and fully in accordance with the maintenance instructions supplied.

Warranty
BiS Valves warrants that for a period of 12 months from the date of first use or for a period of 24 months from date of delivery (whichever expires first) the product shall be free from material defects in design, material and workmanship. Should it be necessary to return a product, please contact the BiS Valves sales team in advance. This is of additional importance if the goods are being returned to the UK from non EU countries, to allow BiS to ensure compliance with the relevant UK customs procedures. Failure to comply can result in a demand for payment of Customs Duties which BiS may have to pass on to the customer.